Step 1: Load the Data Purpose: Import the dataset and inspect its structure. Summary: The dataset contains metadata about Netflix titles including type, cast, country, release year, rating, and duration. We'll begin by cleaning and exploring this data.

In [1]:
```python
import pandas as pd

df = pd.read_csv('netflix_titles.csv')
df.head()
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 12 columns):
 #   Column        Non-Null Count  Dtype 
---  ------        --------------  ----- 
 0   show_id       8807 non-null   object
 1   type          8807 non-null   object
 2   title         8807 non-null   object
 3   director      6173 non-null   object
 4   cast          7982 non-null   object
 5   country       7976 non-null   object
 6   date_added    8797 non-null   object
 7   release_year  8807 non-null   int64 
 8   rating        8803 non-null   object
 9   duration      8804 non-null   object
 10  listed_in     8807 non-null   object
 11  description   8807 non-null   object
dtypes: int64(1), object(11)
memory usage: 825.8+ KB
```

Step 2: Handle Missing Values & Understand Stats Purpose: Identify nulls and get summary statistics. Summary: There are missing values in columns like cast, country, and director. We'll address these later. The dataset contains both categorical and numeric data.

In [2]:
```python
df.isnull().sum()
df.describe(include='all')
```

Out [2]:

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description
count	8807	8807	8807	6173	7982	7976	8797	8807.000000	8803	8804	8807	8807
unique	8807	2	8807	4528	7692	748	1767	NaN	17	220	514	8775
top	s8807	Movie	Zubaan	Rajiv Chilaka	David Attenborough	United States	January 1, 2020	NaN	TV-MA	1 Season	Dramas, International Movies	Paranormal activity at a lush, abandoned prope...
freq	1	6131	1	19	19	2818	109	NaN	3207	1793	362	4
mean	NaN	NaN	NaN	NaN	NaN	NaN	NaN	2014.180198	NaN	NaN	NaN	NaN
std	NaN	NaN	NaN	NaN	NaN	NaN	NaN	8.819312	NaN	NaN	NaN	NaN
min	NaN	NaN	NaN	NaN	NaN	NaN	NaN	1925.000000	NaN	NaN	NaN	NaN
25%	NaN	NaN	NaN	NaN	NaN	NaN	NaN	2013.000000	NaN	NaN	NaN	NaN
50%	NaN	NaN	NaN	NaN	NaN	NaN	NaN	2017.000000	NaN	NaN	NaN	NaN
75%	NaN	NaN	NaN	NaN	NaN	NaN	NaN	2019.000000	NaN	NaN	NaN	NaN
max	NaN	NaN	NaN	NaN	NaN	NaN	NaN	2021.000000	NaN	NaN	NaN	NaN

Step 3: Clean Duration Column & Missing Data Purpose: Split duration into number and type (e.g., minutes or seasons) Summary : The 'duration' column was split into numeric and type columns. Also, missing countries were filled as "Unknown" for consistency.

In [3]:
```python
df['duration_int'] = df['duration'].str.extract(r'(\d+)').astype(float)
df['duration_type'] = df['duration'].str.extract(r'([a-zA-Z]+)')
```

In [4]:
```python
df['country'] = df['country'].fillna('Unknown')
```

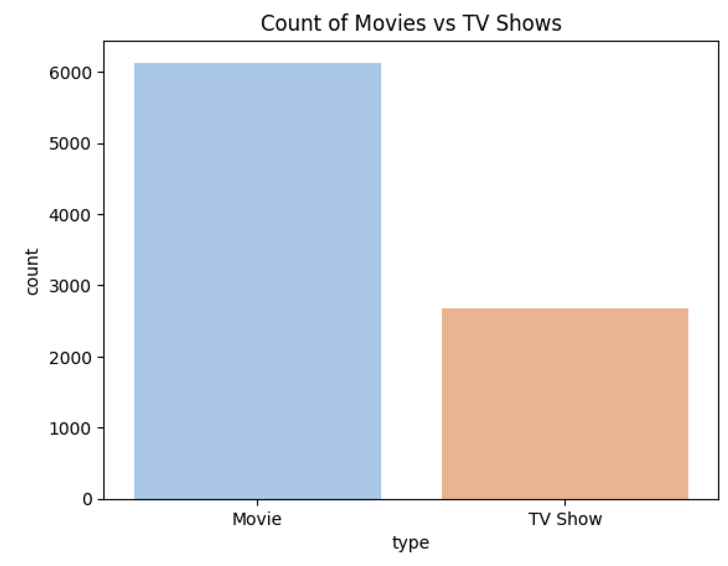
In [5]:
```python
# ✅ Extract numeric duration
df['duration_int'] = df['duration'].str.extract(r'(\d+)').astype(float)
df['duration_type'] = df['duration'].str.extract(r'([a-zA-Z]+)')

# ✅ Fill missing countries safely
df['country'] = df['country'].fillna('Unknown')
```

Step 4: Movies vs TV Shows Count Purpose: Visualize content type distribution. Summary : Netflix has more movies than TV shows. Movies are the dominant content type on the platform.

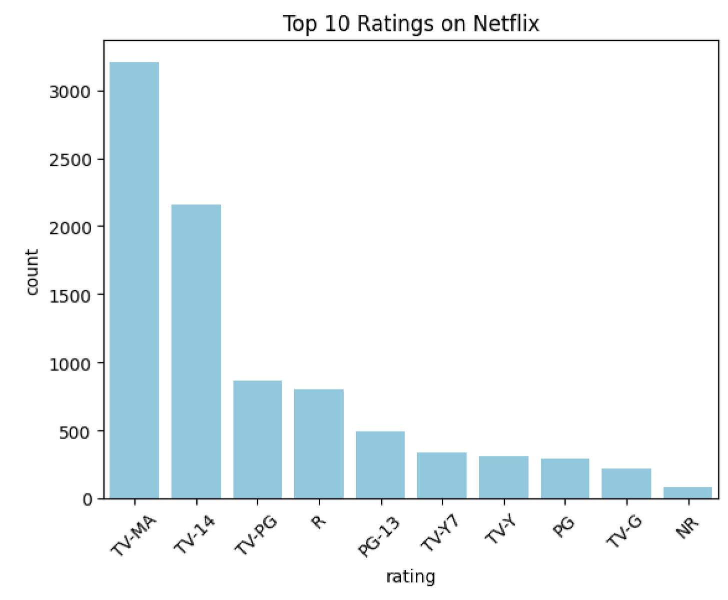
In [7]:
```python
import seaborn as sns
import matplotlib.pyplot as plt

sns.countplot(data=df, x='type', hue='type', palette='pastel', legend=False)
plt.title('Count of Movies vs TV Shows')
plt.show()
```



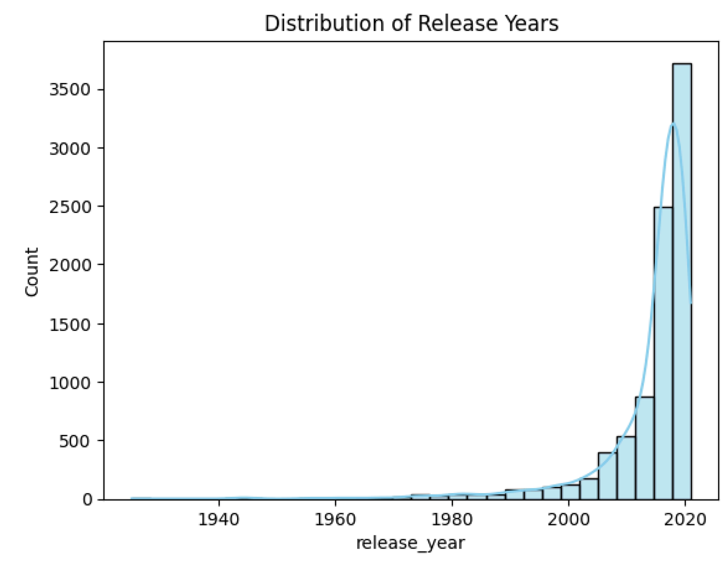
Step 5: Ratings Distribution Purpose: Understand how content is rated. Summary: Most content is rated 'TV-MA' and 'TV-14', suggesting a focus on mature audiences.

In [13]:
```python
sns.countplot(
    data=df,
    x='rating',
    order=df['rating'].value_counts().head(10).index,
    color='skyblue'  # optional single color
)
plt.title('Top 10 Ratings on Netflix')
plt.xticks(rotation=45)
plt.show()
```



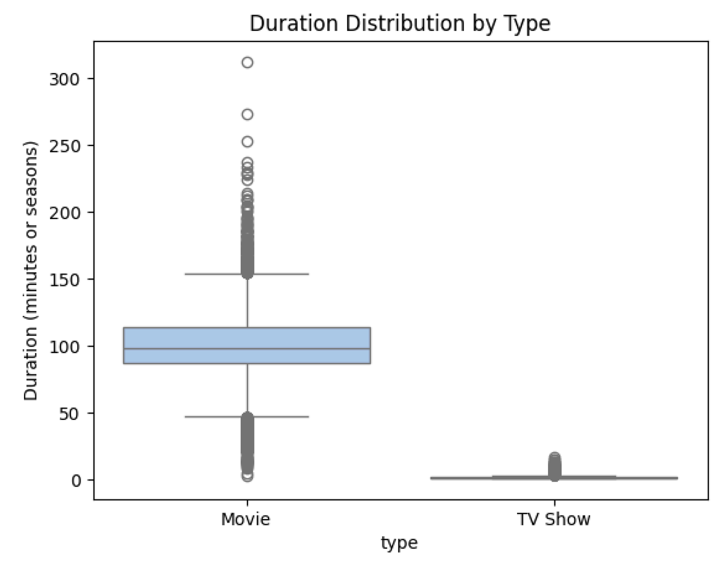
Step 6: Release Year Distribution Purpose: Explore trends over time. Summary: Content releases have grown steadily since 2010, peaking around 2018–2020.

In [14]:
```python
sns.histplot(data=df, x='release_year', bins=30, kde=True, color='skyblue')
plt.title('Distribution of Release Years')
plt.show()
```



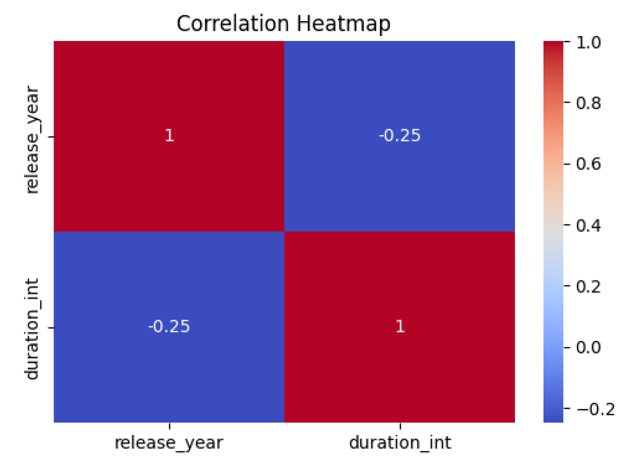
Step 7: Duration Comparison by Type Purpose: Compare movie lengths vs TV show seasons. Summary: Most movies are around 90 minutes long, while TV shows range from 1 to 3 seasons on average.

In [16]:
```python
sns.boxplot(data=df, x='type', y='duration_int', hue='type', palette='pastel', legend=False)
plt.title('Duration Distribution by Type')
plt.ylabel('Duration (minutes or seasons)')
plt.show()
```



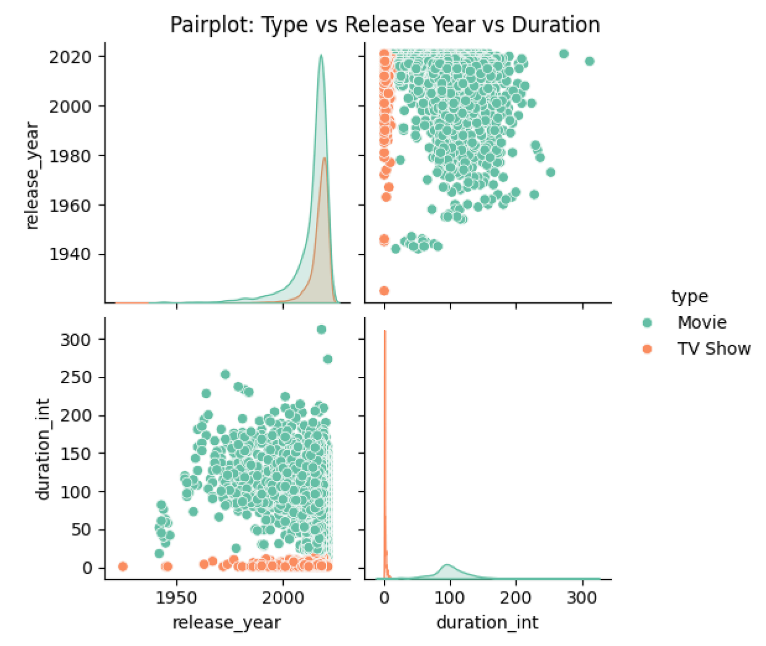
Step 8: Correlation Heatmap Purpose: Check numeric correlations. Summary: There is no strong correlation between release year and content duration.

In [17]:
```python
plt.figure(figsize=(6, 4))
sns.heatmap(df[['release_year', 'duration_int']].dropna().corr(), annot=True, cmap='coolwarm')
plt.title('Correlation Heatmap')
plt.show()
```



Step 9: Pairplot for Multivariate Insight Purpose: Explore relationships across features. Summary: Clear separation between movies and TV shows. TV shows cluster by season count, while movie durations are more varied.

In [18]:
```python
sns.pairplot(df[df['duration_int'].notnull()][['type', 'release_year', 'duration_int']], hue='type', palette='Set2')
plt.suptitle('Pairplot: Type vs Release Year vs Duration', y=1.02)
plt.show()
```



Summary of Findings:

---Netflix offers more Movies than TV Shows.

----Most content is targeted at mature viewers (TV-MA, TV-14).

----Content release surged between 2015–2020.

----Movies average around 90 minutes; TV shows range between 1–3 seasons.

----Duration and release year show no significant correlation.

----The US is the leading content contributor.

In [21]:

```
Requirement already satisfied: pandoc in c:\users\abhis\appdata\local\programs\python\python313\lib\site-packages (2.4)
Requirement already satisfied: plumbum in c:\users\abhis\appdata\local\programs\python\python313\lib\site-packages (from pandoc) (1.9.0)
Requirement already satisfied: ply in c:\users\abhis\appdata\local\programs\python\python313\lib\site-packages (from pandoc) (3.11)
```

```
Requirement already satisfied: pywin32 in c:\users\abhis\appdata\local\programs\python\python313\lib\site-packages (from plumbum->pandoc) (308)
Note: you may need to restart the kernel to use updated packages.
```

In [ ]:

```

```